SignMuon, MuonSign, and the Role of Error Feedback: Theory Addendum

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Abstract

We organize the theory revision around the reviewers' questions. We give a geometric sufficient condition for vanilla SignMuon, extend it to stochastic EMA and Nesterov momentum, and give convex quadratic EF21-SignMuon counterexamples under decreasing steps and finite-budget tuning. Further extensions are explicitly marked as open targets.

Status. This working addendum includes proofs of the statements below. Results inherited from prior work or the existing manuscript are identified as such; open targets are not claimed as theorems. Section 4 reports completed heterogeneous transfer and validation-tuned experiments.

1 Why Can SignMuon Work in Practice?

Critique: pqje, 7S1b, D4r2, Tx34. A worst-case failure does not predict an empirical ranking. The relevant positive question is when signing a polar factor preserves descent. We use the diagonal-isotropy argument of Wang, Shen, and Zhang (2026) and supply a stochastic momentum extension for interacting parameter blocks.

Write $X = (X_\ell)_{\ell=1}^p$, with $X_\ell \in \mathbb{R}^{m_\ell \times n_\ell}$, and set

$$d = \sum_\ell m_\ell n_\ell, \qquad R = \sum_\ell \min(m_\ell, n_\ell),$$
$$\mathcal{N}(G) = \sum_\ell \|G_\ell\|_*.$$

The joint Frobenius norm is $\|X\|_F^2 = \sum_\ell \|X_\ell\|_F^2$. Polar factors are exact throughout this section; zero signs may be resolved to either binary value.

1.1 A sufficient geometric condition

For a nonzero rank-r matrix $H = U\Sigma V^\top$, define

$$Q = UV^\top, \quad S = \operatorname{sign}(Q), \quad a = \operatorname{tr}(\Sigma)/r,$$
$$c = \operatorname{diag}(U^\top S V), \quad b = \|Q\|_1/r,$$
$$\varepsilon(H) = \frac{\|c - b\mathbf{1}\|_2}{b\sqrt{r}}, \qquad \rho(H) = \frac{\|\Sigma - aI\|_F}{a\sqrt{r}}.$$

Here $\|\cdot\|_1$ is entrywise. Each c_j measures how the signed polar factor correlates with the singular direction $u_j v_j^\top$. The quantity ε measures how unequal these correlations are relative to their positive mean; ρ measures how unequal the singular values are relative to their mean.

Lemma 1 (Geometric alignment) *With the notation above,*

$$\langle H, S\rangle \geq a\|Q\|_1\big(1 - \varepsilon(H)\rho(H)\big). \tag{1}$$

Consequently, if $\varepsilon(H)\rho(H) \leq 1 - \delta$ for $\delta \in (0, 1]$, then $\langle H, S\rangle \geq \delta\|H\|_*$.

Proof. The first bound is the cancellation argument in OLion, Lemma 4.3:

$$\langle H, S\rangle = a\|Q\|_1 + \sum_j (\sigma_j - a)(c_j - b).$$

Cauchy–Schwarz bounds the last term below by $-a\|Q\|_1\varepsilon(H)\rho(H)$. For the consequence, $\|Q\|_1 \geq \|Q\|_* = r$ and $ar = \|H\|_*$. □

Assumption 1 (Smoothness and noise) *The objective is bounded below by f_* and its gradient is L-Lipschitz in the joint Frobenius norm. Conditional on the history before query t,*

$$\mathbb{E}_t G_t = \nabla f(X_t), \qquad \mathbb{E}_t\|G_t - \nabla f(X_t)\|_F^2 \leq \sigma^2.$$

Assumption 2 (Uniform geometric margin) *For a fixed $\delta \in (0, 1]$, independent of the horizon, every nonzero block of the actual momentum signal H_t satisfies, almost surely,*

$$\varepsilon(H_{t,\ell})\rho(H_{t,\ell}) \leq 1 - \delta \qquad \textit{for all } t, \ell.$$

Relation to OLion. The condition says that correlation imbalance times singular-value spread must stay below one by a fixed amount $\delta > 0$. Thus a more uneven spectrum requires tighter balance across singular directions. The lemma then guarantees positive alignment of at least $\delta\|H\|_*$. Assumption 2 is our quantitative adaptation, not a verbatim assumption from OLion. Their Assumption 4.2 bounds ε; their descent measure (11) contains $1 - \varepsilon\rho$, and the discussion after Theorem 4.4 adds spectral-spread and density conditions. We make a uniform positive margin explicit on each momentum block. This is an adaptation of their geometric requirement, not a new geometric principle.

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The margin excludes our ascent witness by Lemma 1; small isotropy error alone is insufficient. No separate density or spectral-flatness condition is needed for the dimension-dependent bounds below. Joint smoothness allows cross-layer coupling, unlike the manuscript's block-only smoothness condition.

1.2 Stochastic momentum convergence

Initialize $M_0 = 0$. With $\beta = 1 - \mu \in (0, 1]$ and $\theta = 0$ for EMA or $\theta = \beta$ for Nesterov, run

$$\begin{aligned} M_{t+1} &= \mu M_t + \beta G_t, \\ H_t &= (1-\theta) M_{t+1} + \theta G_t, \\ S_{t,\ell} &= \operatorname{sign}(\operatorname{polar}(H_{t,\ell})), \\ X_{t+1} &= X_t - \eta S_t. \end{aligned} \tag{2}$$

This is vanilla SignMuon, without error feedback or weight decay. Let $\Delta = f(X_0) - f_*$ and $I_0 = \|\nabla f(X_0)\|_F$.

Theorem 1 (Conditional convergence of SignMuon) *Under Assumptions 1–2, the iterates of* (2) *satisfy*

$$\frac{1}{T} \sum_{t<T} \mathbb{E}\mathcal{N}(\nabla f(X_t)) \le \frac{\Delta}{\delta \eta T} + \frac{L d \eta}{2\delta} + K_\delta B_T, \tag{3}$$

where $K_\delta = \sqrt{R} + \sqrt{d}/\delta$ *and*

$$B_T = (1-\theta) \left[\frac{\mu I_0}{\beta T} + \frac{\mu L \eta \sqrt{d}}{\beta} + \sigma \sqrt{\frac{\beta}{2-\beta}} \right] + \theta \sigma.$$

For fixed dimensions and problem constants, choosing $\beta = T^{-1/2}$ *and* $\eta = \eta_0 T^{-3/4}$*, with fixed* $\eta_0 > 0$*, gives an* $\mathcal{O}(T^{-1/4})$ *bound for either momentum rule. With exact gradients and no momentum,* $\eta = \eta_0 T^{-1/2}$ *gives* $\mathcal{O}(T^{-1/2})$.

Proof. Put $g_t = \nabla f(X_t)$. Since $\|S_t\|_F \le \sqrt{d}$ and $\mathcal{N}(A) \le \sqrt{R}\|A\|_F$, Lemma 1 gives

$$\begin{aligned} \langle g_t, S_t \rangle &\ge \delta \mathcal{N}(H_t) - \sqrt{d} \|H_t - g_t\|_F \\ &\ge \delta \mathcal{N}(g_t) - (\delta \sqrt{R} + \sqrt{d}) \|H_t - g_t\|_F. \end{aligned}$$

Joint smoothness and telescoping therefore give (3) with B_T replaced by $T^{-1} \sum_{t<T} \mathbb{E}\|H_t - g_t\|_F$.

To bound this error, set $\xi_t = G_t - g_t$. Expanding the EMA yields

$$\begin{aligned} M_{t+1} - g_t = & -\mu^{t+1} g_0 + \sum_{s=1}^{t} \mu^{t-s+1} (g_{s-1} - g_s) \\ & + \beta \sum_{s=0}^{t} \mu^{t-s} \xi_s, \end{aligned}$$

The middle term has norm at most $\mu L \eta \sqrt{d}/\beta$. Conditional centering makes distinct noise terms orthogonal in expectation, so the last term has expected norm at most $\sigma\sqrt{\beta/(2-\beta)}$. Averaging $\mu^{t+1} I_0$ gives at most $\mu I_0/(\beta T)$. Finally, $H_t - g_t = (1-\theta)(M_{t+1} - g_t) + \theta \xi_t$. The triangle inequality gives B_T; substitution gives both rates. □

Interpretation. The theorem supplies a sufficient mechanism for successful training, not a ranking against the EF21 variants. OLion reports small isotropy errors for selected GPT-2 matrices; this motivates checking the full product $\varepsilon\rho$ on our momentum signals. Neither smoothness nor the geometric margin is asserted automatically for a neural network. The rate controls a first moment; squaring it does not bound the expected squared gradient norm.

2 How General Are the Divergence Results?

Critique: Tx34, D4r2. The plain sign placements fail under arbitrary positive schedules. A separate convex quadratic construction makes EF21-SignMuon fail under decreasing steps and rules out uniform rates from horizon-tuned constant learning rates and momentum.

2.1 Lower-bounded examples for the plain methods

Let s be any of the three direction maps in the paper: $\operatorname{sign} \circ \operatorname{polar}$, $\operatorname{polar} \circ \operatorname{sign}$, or $\operatorname{sign} \circ \operatorname{polar} \circ \operatorname{sign}$. Each has a witness $G \neq 0$ with $a := \langle G, s(G) \rangle < 0$ and no relevant sign ties.

Proposition 1 (Arbitrary schedules) *For each such witness, define*

$$f(X) = \log(1 + \exp(\langle G, X \rangle)).$$

This function is convex, bounded below, and globally smooth with $L \le \|G\|_F^2/4$*. From* $X_0 = 0$*, for any positive step sizes and any momentum coefficients* $\mu_t \in [0, 1)$*, either EMA or Nesterov generates*

$$X_t = -s(G) \sum_{k<t} \eta_k, \qquad \|\nabla f(X_t)\|_F \ge \tfrac{1}{2} \|G\|_F.$$

If $\sum_k \eta_k = \infty$*, then* $f(X_t) \to \infty$*. The objective is independent of the schedule and the horizon.*

Proof. Writing $z = \langle G, X \rangle$ gives $\nabla f(X) = q(z) G$, where $q(z) = (1 + e^{-z})^{-1} > 0$. Every momentum signal is a positive multiple of G, so positive homogeneity of s fixes the direction at $s(G)$. Thus $\langle G, X_t \rangle = -a \sum_{k<t} \eta_k \ge 0$, which implies $q(\langle G, X_t \rangle) \ge 1/2$ and the stated divergence. The Hessian is $q(z)(1 - q(z)) \operatorname{vec}(G) \operatorname{vec}(G)^\top$, proving convexity and smoothness. □

2.2 EF21-SignMuon: decreasing steps and finite budgets

The fixed-step construction scales its gradient floor with the learning rate. The following example instead keeps the smoothness, initial gap, dimension, and gradient floor uniform. Failure means nonstationarity; the objective need not increase without bound.

Lemma 2 (An invisible gradient component) *There are* $A, B \in \mathbb{R}^{8\times 8}$ *and* $S \in \{\pm 1\}^{8\times 8}$ *such that*

$$\begin{gathered} \|A\|_F = 1, \quad \|B\|_F^2 = 7, \quad \langle A, B \rangle = \langle A, S \rangle = 0, \\ \operatorname{polar}(aA + bB) = A + \operatorname{sign}(b) B \quad (a > 0,\ b \neq 0), \\ \operatorname{sign}(A + B) = S, \qquad \operatorname{sign}(A - B) = -S. \end{gathered}$$

Every entry of $A+B$ and $B-A$ has magnitude greater than $1/50$. Put $h=\langle B,S\rangle/64>0$ and $\kappa=64h^2\le 8$. For alternating targets $D_t=A+(-1)^{t-1}B$, scaled-sign EF21, initialized at $E_0=0$, produces

$$E_t=E_{t-1}+C(D_t-E_{t-1})=(-1)^{t-1}hS.$$

In particular, $\langle A,E_t\rangle=0$ for every t.

The algebraic certificate below supplies the matrices. For the recursion, the two strict sign identities imply $S_{ij}B_{ij}>|A_{ij}|$ entrywise. The first message is hS. If $E_{t-1}=-shS$ and $D_t=A+sB$, their residual has signs sS and mean modulus $2h$, so $E_t=shS$.

Theorem 2 (Uniform failure of stationarity) *For every $b\in\mathbb{R}$, put $a_b=\sqrt{1+1/[7(1+b^2)]}$ and define the convex quadratic*

$$f_b(X)=\frac{(a_b+\langle A,X\rangle)^2}{2}+\frac{(\langle B,X\rangle-b)^2}{14(1+b^2)}. \tag{4}$$

It is globally 1-smooth in Frobenius norm, attains its minimum 0, and satisfies $f_b(0)=4/7$. Run exact-gradient, exact-polar EF21-SignMuon from $X_0=M_0=E_0=0$, with either EMA or Nesterov and fixed $\mu\in[0,1)$. Let $r_t>0$ satisfy $r_t>\mu r_{t-1}$ for $t\ge 2$, and set

$$\eta_t=r_t+r_{t+1},\qquad b=-\kappa r_1.$$

Then, for every $t\ge 0$,

$$X_t=-hS\sum_{k=1}^{t}(-1)^{k-1}\eta_k,$$
$$\langle B,X_t\rangle-b=(-1)^t\kappa r_{t+1},$$
$$\|\nabla f_b(X_t)\|_F\ge 1.$$

The same statement holds through any finite horizon when the hypotheses hold only for the required finite sequence.

Proof. The Hessian is the sum of two orthogonal rank-one operators with eigenvalues 1 and $1/(1+b^2)$. A minimizer is $-a_bA+(b/7)B$, and $f_b(0)=a_b^2/2+b^2/[14(1+b^2)]=4/7$. In the article's spectral/nuclear smoothness convention, it also satisfies $\|\nabla f_b(X)-\nabla f_b(Y)\|_*\le 8\|X-Y\|_{2\to 2}$, since $\|A\|_*=1$ and $\|B\|_*=7$. Suppose the claimed trajectory holds through X_{t-1}. Its gradient is

$$G_t=a_bA+(-1)^{t-1}\frac{\kappa r_t}{7(1+b^2)}B.$$

The EMA coefficient of A is $a_b(1-\mu^t)>0$. After removing the sign $(-1)^{t-1}$ and a positive common factor, its coefficient of B is

$$r_t-\mu r_{t-1}+\mu^2 r_{t-2}-\cdots+(-\mu)^{t-1}r_1>0.$$

Pair adjacent terms; each pair is positive by the hypothesis, as is the unpaired last term. Consecutive EMA coefficients of B therefore have opposite signs. Nesterov's signal $(1+\mu)M_t-\mu M_{t-1}$ has the same B sign and a positive A coefficient as well. Lemma 2 gives D_t and E_t. The update $X_t=X_{t-1}-\eta_tE_t$ and $\eta_t=r_t+r_{t+1}$ complete the induction. Finally, $\langle A,X_t\rangle=0$ implies $\langle A,\nabla f_b(X_t)\rangle=a_b\ge 1$, proving the gradient floor. □

Decreasing steps. With $\mu=0$, every strictly decreasing positive schedule $\eta_t\to 0$ is covered: take $r_t=\sum_{k=t}^{\infty}(-1)^{k-t}\eta_k$. The alternating-series bounds give $0<r_t<\eta_t$ and $r_t+r_{t+1}=\eta_t$. Thus one fixed quadratic defeats the entire run, including $\eta_t=a/t$ and $a/\sqrt{t}$. The iterates converge to a point whose gradient is a_bA, of norm at least 1. For any fixed $\mu<1$, choose $a>0$, $0<p\le 1$, and $\tau\ge 0$ large enough that

$$r_t=\frac{a}{(t+\tau)^p},\qquad \left(\frac{1+\tau}{2+\tau}\right)^p>\mu.$$

Both momentum variants are then covered. These steps satisfy $\eta_t\asymp t^{-p}$ and $\sum_t\eta_t=\infty$; choosing $p>1/2$ also gives $\sum_t\eta_t^2<\infty$.

Parameters chosen for a finite budget. For every horizon T, any constant $\eta_T>0$ and any fixed $\mu_T\in[0,1)$, choose $r_t=\eta_T/2$ and $b_T=-\kappa\eta_T/2$. Then

$$\min_{0\le t\le T}\|\nabla f_{b_T}(X_t)\|_F\ge 1,$$
$$L_F=1,\qquad f_{b_T}(0)-\inf f_{b_T}=4/7.$$

Thus no uniform vanishing stationarity rate over this smooth, lower-bounded class follows by tuning a constant learning rate and momentum to T. The dimension and compressor are also fixed, and the noise is zero. With $\mu=0$, any finite nonincreasing positive schedule is covered: set $r_{T+1}=\eta_T/2$ and recursively $r_t=\eta_t-r_{t+1}>0$. These are uniform worst-case statements, with the instance allowed to depend on the chosen schedule; arbitrary varying momentum is not asserted.

Why contraction does not suffice. Here $\kappa\simeq 5.039$. The achieved scaled-sign contraction is $\kappa/8\simeq 0.630$ initially and $4\kappa/(8+3\kappa)\simeq 0.872$ thereafter. Nevertheless, every estimator is orthogonal to the fixed gradient component a_bA. All finite-time momentum signals have full rank and all compressed residuals have strict signs. As the steps decrease, the gradients approach rank one while their polar factors keep alternating. Smoothness of the objective therefore does not make the polar target slowly varying.

Exact algebraic certificate. The following data make Lemma 2 explicit. Let K be the skew-symmetric integer matrix

$$K=\begin{pmatrix}
0 & -6 & -10 & 15 & 4 & -3 & -9 & 3\\
6 & 0 & -3 & -7 & 0 & -12 & -4 & 0\\
10 & 3 & 0 & 20 & 0 & 8 & -1 & 4\\
-15 & 7 & -20 & 0 & -21 & -7 & 0 & -1\\
-4 & 0 & 0 & 21 & 0 & -6 & -15 & 7\\
3 & 12 & -8 & 7 & 6 & 0 & -17 & 0\\
9 & 4 & 1 & 0 & 15 & 17 & 0 & 12\\
-3 & 0 & -4 & 1 & -7 & 0 & -12 & 0
\end{pmatrix},$$

and put $D=(20I-K)(20I+K)^{-1}$, $S=\operatorname{sign}(D)$, $w=(-1,-8,-9,0,0,-12,7,-8)^\top$. Let $z=w+\xi e_6$, where $\xi\in(251/250,201/200)$ is the unique root of

$$3903331846\xi^2+1548933728\xi-5493727089=0.$$

Define

$$A = \frac{Dzz^\top}{z^\top z}, \qquad B = D - A.$$

Skew-symmetry gives $D^\top D = I$; the polynomial is a nonzero multiple of $z^\top D^\top S z$, so $\langle A, S\rangle = 0$. The rank-one orthogonal projector $zz^\top/(z^\top z)$ gives the stated norms and polar identity. Rational interval arithmetic on the displayed interval verifies, entrywise,

$$S_{ij}D_{ij} > 1/50, \qquad S_{ij}(D - 2A)_{ij} > 1/50.$$

For reproducibility, the accompanying script `research/ef21_signmuon_vanishing_steps.py` verifies these inequalities with exact fractions: after multiplying the second by $z^\top z$, each is a quadratic inequality checked at the interval endpoints and, when applicable, its vertex. It also checks root existence, uniqueness, and orthogonality exactly; no numerical search is part of the certificate.

3 Which Guarantees Cover the Implementations?

Critique: pqje, Ueq5, D4r2, 7S1b. The new SignMuon theorem covers stochastic EMA and Nesterov under joint smoothness. Other implementation differences require distinct arguments.

EF21 momentum and schedules: existing revision. The revised appendix already supplies Nesterov and deterministic-schedule bounds for EF21-MuonUSign and EF21-MuonSign. For constant or cosine learning rates, their form is

$$\frac{\sum_{t<T}\eta_t \mathbb{E}\mathcal{N}(\nabla f(X_t))}{\sum_{t<T}\eta_t} \le \frac{\Delta + \bar\eta A_\beta}{\sum_{t<T}\eta_t} + \bar\eta B_\beta + V_\beta,$$

with fixed momentum within the run, exact polar factors, and explicit initialization, movement, and noise constants. Fixed peak learning rate and fixed noise leave a residual in this bound. The gradient is evaluated at the server model X_t, including for the compressed downlink. These results still use the manuscript's restrictive block-only smoothness.

EF21 cross-layer coupling: open target. Extend the tracking estimates to joint Frobenius smoothness of each local objective. The required change is to control each local gradient increment by the displacement of the entire model. The current block-separable proofs do not establish this extension. First- and second-moment guarantees must remain separate.

Finite orthogonalization and local voting. Theorem 1 concerns signing the polar factor of one gradient estimator. A full-precision average followed by this update is different from voting on locally signed polar factors. Both finite-solver and voting effects can be exposed as a defect relative to the ideal S_t:

$$a_t = \max\{0, \langle H_t, S_t - \widehat{S}_t\rangle\}, \qquad \|\widehat{S}_t\|_F \le \sqrt{d}.$$

For the run using $\widehat{S}_t$, the same proof adds $\frac{1}{\delta T}\sum_{t<T}\mathbb{E}a_t$ to (3), provided the stated assumptions hold along that run. For binary updates,

$$a_t \le 2 \sum_{(\ell,i,j)\in\mathcal{F}_t} |(H_{t,\ell})_{ij}|,$$

where $\mathcal{F}_t$ is the set of entries whose signs differ. Bounding this weighted disagreement for the actual solver or aggregation rule remains necessary; a small error before signing alone is insufficient. No conclusion about non-IID local training follows from the centralized theorem.

4 Does Heterogeneity Change the Empirical Ranking?

Critique: 7S1b, pqje, D4r2. Table 1 compares the existing homogeneous results with two CNN2/CIFAR-10 experiments under Dirichlet label skew, concentration $\alpha = 0.1$. Both retain 11 fully participating clients, 2000 rounds, cosine scheduling, momentum $\mu = 0.9$, unit-gain scaling, zero weight decay, and auxiliary AdamW learning rate 0.001. Each round accumulates three minibatches of up to 64 images per client at the broadcast model, without local model steps. Client objectives have equal weight; initialization and partitions are paired across methods at each seed.

Experiment 1: transfer homogeneous settings. Keep each method's homogeneous learning rate and momentum $\mu = 0.9$. All ten article algorithms run together (50 final runs). Changing only the client partition measures robustness without retuning.

Experiment 2: validation-based learning-rate tuning. Use the homogeneous 1–2–5 learning-rate grid and initial search centers, keeping momentum fixed. Start with five configurations per method; a boundary winner adds two rates, with at most three base extensions. MuonSign still selected the lower endpoint after those extensions, so we tested two further rates, 10^{-5} and 2×10^{-5}. Its selected rate remains 5×10^{-5} and is now interior to the grid. All ten methods have interior validation winners.

The completed search contains 72 validation runs, each trained for 2000 rounds with tuning seed 0. Select only by mean accuracy over the last five validation evaluations, holding out 5000 images before partitioning the remaining 45,000; test accuracy does not enter selection. Final runs retrain on all 50,000 images. Transfer and tuned comparisons each contain ten methods and five final seeds. Muon's selected rate matches its transfer rate, so its five final runs are shared: the two comparisons use 95 unique final runs. Both experiments report the last-five test mean as mean $\pm$ sample standard deviation over paired seeds $0, \ldots, 4$. The retuned learning rates test whether the heterogeneous ranking survives learning-rate selection at the same fixed momentum $\mu = 0.9$.

The heterogeneous ranking persists after validation-based learning-rate tuning. **EF21-MuonSign reaches** $84.03 \pm 0.52\%$ **and EF21-MuonUSign** $83.84 \pm 0.32\%$; they are the only methods above 80% in the tuned comparison. SGD reaches $79.71 \pm 2.03\%$ and Adam $75.16 \pm 1.07\%$. Retuning improves SignMuon from 35.92% to 42.96% and output-level EF21-SignMuon from 45.46% to 49.73%, while Muon remains at 46.02%. Thus the advantage of gradient-side error feedback under this label skew survives learning-rate selection. The contrast with SignMuon's leading compressed

Optimizer	Homogeneous reference		Experiment 1: transfer	Experiment 2: LR retuned	
	η_0	Test acc. (%)	Test acc. (%)	η_0	Test acc. (%)
SignMuon	0.1	85.72 ± 0.24	35.92 ± 8.49	0.005	42.96 ± 5.94
EF21-MuonUSign	0.01	83.56 ± 0.41	$\mathbf{82.99 \pm 0.46}$	0.02	$\mathbf{83.84 \pm 0.32}$
EF21-MuonSign	0.05	83.99 ± 0.28	$\mathbf{83.23 \pm 0.36}$	0.02	$\mathbf{84.03 \pm 0.52}$
EF21-SignMuon	0.02	84.71 ± 0.15	45.46 ± 7.66	0.2	49.73 ± 6.69
Muon	0.1	85.98 ± 0.26	46.02 ± 8.55	0.1	46.02 ± 8.55
MuonUSign	0.05	84.56 ± 0.35	26.21 ± 2.75	0.001	44.23 ± 3.28
MuonSign	0.02	82.94 ± 0.19	25.63 ± 3.98	5×10^{-5}	40.75 ± 0.98
SignSGD	0.01	81.44 ± 0.15	26.42 ± 3.23	0.0005	36.24 ± 4.64
SGD	0.05	81.69 ± 0.37	81.03 ± 0.62	0.1	79.71 ± 2.03
Adam	0.001	77.12 ± 1.04	71.17 ± 7.16	0.0005	75.16 ± 1.07

Table 1: CNN2/CIFAR-10, 11 clients, 2000 rounds, five seeds. Both new experiments use Dirichlet(0.1) label skew. Experiment 1 retains the reference learning rates; Experiment 2 selects learning rates using a held-out 5000-image validation set and tuning seed 0. Momentum stays $\mu = 0.9$ throughout. Accuracy is mean $\pm$ sample standard deviation of per-seed last-five test means over five final seeds. Both comparisons are complete.

accuracy on IID data identifies when the placement of error feedback changes the useful algorithm.

5 What Is New, and What Must Be Corrected?

Critique: Tx34, pqje, 7S1b. The alignment lemma is credited to OLion. The contribution developed here is its explicit stochastic EMA/Nesterov bound under joint smoothness, together with the bounded-below arbitrary-schedule counterexamples and the convex quadratic EF21-SignMuon obstruction with uniform constants under decreasing steps and finite-budget tuning. Verification of the geometric margin on our neural-network trajectories remains future work.

The baseline EF21-MuonUSign/MuonSign guarantees instantiate Gruntkowska et al. (2025). The Nesterov and deterministic-schedule bounds supply additional refinements.

The updated SignMuon article now proves conditional guarantees for the actual signed vote, including a stochastic momentum result (Mishra et al. 2026). The earlier description of its theory as covering only SignSGD must be replaced. Its alignment assumptions and our geometric sufficient condition are compatible with worst-case divergence.

Frobenius contractivity suffices for the EF21 guarantees. Their first- and second-moment rates use separate proofs and tuning. These guarantees characterize convergence conditions; experiments determine the empirical ranking.

5.1 Draft rebuttal points

Each reviewer block below fits 1,500 characters, including the quotations, spaces, and LaTeX markup. Quotations are exact excerpts from the reviews. Responses cite the revised manuscript and the results proved in this addendum.

Reviewer Tx34. *"Fixed-step divergence does not rule out a finite-time convergence rate."* **Response.** A new convex quadratic EF21-SignMuon example keeps Frobenius $L = 1$, initial gap $4/7$, and gradient norm at least 1 at every iterate. It defeats any horizon-tuned constant learning rate and fixed EMA/Nesterov momentum. It also covers arbitrary strictly decreasing steps without momentum, and inverse-power schedules with either fixed momentum variant. The proof includes an exact algebraic certificate.

"The theory does not predict the experimental ranking." **Response.** A sufficient condition is balanced correlation of the signed polar factor with momentum's singular directions, with tighter balance for more uneven singular values. Under this OLion-based condition, joint smoothness and bounded gradient noise, the average expected gradient norm is $O(T^{-1/4})$ for EMA/Nesterov with the stated tuning. This gives a measurable criterion separating our ascent examples from trajectories admitting convergence guarantees.

"Sufficient conditions are treated as necessary conditions." **Response.** Frobenius contractivity of scaled sign is sufficient to apply EF21-Muon theory under its stated assumptions and tuning. Spectral contractivity is a sufficient tool in the compared analysis, not a necessary property of convergent compressors.

Reviewer pqje. *"under what reasonable assumptions does sign compression after orthogonalization preserve a useful descent direction"* **Response.** OLion's condition supplies a concrete answer: correlations of the signed polar factor with momentum's singular directions must be sufficiently balanced relative to singular-value spread. Under this condition at every step, joint smoothness and bounded gradient noise, our tuned EMA/Nesterov theorem gives an $O(T^{-1/4})$ bound on the average expected gradient norm. Correlation imbalance and spectral spread are measurable along a training trajectory.

"The nanoGPT experiments do not actually evaluate the proposed distributed compression methods." **Response.** NanoGPT tests the update rule after full-precision gradient averaging. Federated CIFAR-10 tests locally compressed messages and aggregation, with per-round uplink and downlink costs reported separately.

"Are any experiments covered by the convergence guar-

antees?" **Response.** With exact polar factors and the stated block-smoothness/noise assumptions, the revised EF21 bounds cover fixed EMA/Nesterov momentum and constant/cosine schedules. They give an explicit transient plus step-size and noise terms; the prescribed horizon tuning makes all terms vanish.

Reviewer Ueq5. *"the layer-wise smoothness assumption forces separability across layers."* **Response.** The new vanilla SignMuon theorem uses joint Frobenius smoothness, so changing one layer may affect every gradient block. Together with the OLion-based geometric condition and bounded gradient noise, this suffices for stochastic EMA/Nesterov convergence on coupled objectives. The existing EF21 guarantees retain their original smoothness assumptions.
"This does not follow by squaring the first-moment bound." **Response.** EF21's $O(T^{-1/4})$ expected-norm and $O(T^{-1/2})$ expected-squared-norm guarantees have separate proofs and tuning. The latter uses gradient-scaled steps. For constant/-cosine schedules, the revised bound explicitly separates transient, step-size, and noise terms.
"The authors should identify the correct results and revise the affected comparisons." **Response.** The revised main table and normalization ablation both report MuonSign $82.94 \pm 0.19\%$ and SignSGD $81.44 \pm 0.15\%$ over five seeds. MuonSign is therefore 1.50 percentage points higher. SignMuon remains ahead at $85.72 \pm 0.24\%$.

Reviewer D4r2. *"Ideally, the paper would provide a resolution to the tension it has documented."* **Response.** A sufficient condition, adapted from OLion, balances the signed polar factor's correlations across momentum's singular directions relative to singular-value spread. It guarantees descent for the momentum linear model and, with joint smoothness and bounded gradient noise, a tuned EMA/Nesterov convergence rate. Our ascent witness violates this measurable condition. The addendum also gives smooth, convex, lower-bounded counterexamples for the plain methods, satisfying Assumption 1.
"the scale of the input is also communicated with each update." **Response.** The communication-accounting appendix already includes one sign bit per matrix entry, a 32-bit scale per EF21 block, and 32-bit auxiliary parameters. On CNN2 this gives 1.0871 bits per parameter per compressed channel per round, a 29.4-fold reduction. EF21-MuonSign compresses both channels; EF21-SignMuon and EF21-MuonUSign retain full-precision downlinks.

Reviewer 7S1b. *"The central theory–practice gap remains largely unresolved."* **Response.** A sufficient condition is balanced correlation of the signed polar factor across momentum's singular directions, with tighter balance for more uneven singular values. Combined with joint smoothness, bounded gradient noise, and the stated tuning, it yields stochastic EMA/Nesterov convergence. This OLion-based criterion is measurable on training trajectories and excludes our ascent witness.
"What parts of the convergence analysis for EF21-MuonUSign and EF21-MuonSign are genuinely new relative to the existing EF21-Muon framework?" **Response.** Based on your review, we proved EF21-MuonUSign and EF21-MuonSign convergence for fixed EMA/Nesterov and constant/cosine schedules, and verified the proofs in Lean. Our separate SignMuon theorem permits interacting layers under joint smoothness and a uniform alignment condition.
"How broadly should the federated-learning conclusions be interpreted?" **Response.** We completed validation LR tuning on CNN2/CIFAR-10 with 11 Dirichlet(0.1) clients, 2000 rounds, five final seeds and no local steps. EF21-MuonSign reaches $84.03 \pm 0.52\%$ and EF21-MuonUSign $83.84 \pm 0.32\%$, versus Muon $46.02 \pm 8.55\%$ and SignMuon $42.96 \pm 5.94\%$. Gradient-side EF still leads after tuning.

References

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